# ELECTROACOUSTIC MUSIC, ORGANIC SOUNDS AND NETWORKS.

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**ABSTRACT**

This paper outlines my PhD project which draws together current and new music practises by writing music that combines live electronic and acoustic performers with new technology in virtual and real performance spaces. The investigation is inspired by Weinberg's (2005) Interconnected Music Networks and details the aesthetic and creative issues that are considered when collaborating with geographically dispersed musicians. This is an interesting area to research in due to its wide focus and the rapid growing interest in the area of Networked Music. The aim of the project is to develop awareness of new performance spaces while creating music suited to this genre that fulfils the composer's aesthetic and practical considerations while using a wide range of sound and instrument sources.

## 1. BACKGROUND

### 1.1. The Composers Aesthetic

*"The measure of a work of art is whether one can sense in it the presence of the artist's body…the corporeal presence of the artist emanating from the work."* Ostertag (2002)
*"When you give a performance that takes your body (or the audience) out of the mundane and into something extraordinary though art, it has a profound appeal."* Ostertag (2002)
*"Listeners read into the piece as a reflection of the personality of the composer."* (Stead 2009)
This paper is part of the author's Creative Phd at The University of Waikato. These are the considerations and aesthetics kept in mind while planning and writing music:

To have a clear sense of intentionality in the style, meaning and purpose of the music.

To appreciate works in which, despite less than perfect performances, the robustness of gesture and intention is clear.

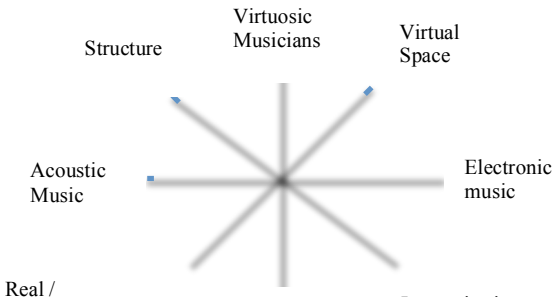
To write music that engages people intellectually and emotionally and provokes a thoughtful response.

To consider at all times the intellectual framework, the idea and its purpose.

To draw from a wide variety of sound sources including natural landscapes and people and utilize a wide range of traditional and non-traditional instruments played by virtuosic and amateur performers.

## 2. INTRODUCTION

*"The world of the acoustic instrument has changed profoundly in the wake of the onslaught of electronic sound and media. How does the instrumentalist (or composer) respond – literally and musically to this."* (Emmerson 2009)
This aim of this research is to write live electroacoustic music for the genre of Networked Music. The reason for this is the interest in the growing genre and the ability to apply well researched acousmatic and acoustic composition techniques and practises to this music. There are eight interlocking focus areas involved in this research.




The eight focus areas of the research question.

The research aims to synthesise these eight areas while exploring the new performance spaces offered by advances in high speed internet technology and computers. The exploration will be a combination of notated works written or directed by the composer as well as mixtures of collaborations with geographically dispersed musicians and composers.

The research will tailor writing music for the new performance space of Networked Music and draw effective practises from the tradition of acousmatic and acoustic music performance – practises such as the blending of acoustic music chamber ensembles or the effective combination of live instruments and electronics, the technology and software for the creation of acousmatic and live electroacoustic music. The

difference will be that this will be performed live over online networks. The performance space in this project is either defined as physical - "natural perception of an environment" or telepresence - "mediated perception of an environment" for example a 'distance space viewed through a video camera'. (Steuer 1992)

### 2.1. Concert Space

Attending live music concerts is a long tradition seeped in history. The architecture of concert halls has been perfected to house classical music, with research and building technique honed to create ideal acoustics.[1] Classical music concerts showcase virtuosity - it is the immediacy of hearing and watching the music performed live that makes it worthwhile to attend.

In the last five years, due to the increase in computer technology, it has become possible to collaborate online with musicians overseas via high speed networks in both real and virtual space.[2] Composers can choose how to use this space and have begun to create a new genre of networked electroacoustic music.

*"With no real physical space users have no way to encounter each other as would a chamber ensemble Therefore participants are aware of each other and intuitively recognize opportunities for interaction by simulating the real world patterns of the performance of chamber music and borrow skills learnt while playing in that medium, of awareness and interaction."* (Föllmer 2005)

Telepresence is a term coined by Marvin Minsky in 1980; "telepresence is defined as the experience of presence in an environment, by means of a communication medium." (Steuer 1992) The research will draw on this approach during the performance of this music.

*"However, when perception is mediated by a communication technology, one is forced to perceive two separate environments simultaneously: the physical environment in which one is actually present, and the environment presented via the medium. The term "telepresence" can be used to describe the precedence of the latter experience in favour of the former; that is, telepresence is the extent to which one feels present in the mediated environment, rather than in the immediate physical environment."* (Steuer 1992)

The Syneme Network is a high speed network set up at the University of University of Calgary, Canada in partnership with the University of Waikato. The network "Supports a new program of artistic practice enabled by high speed fibre optic research links that afford multiple artist/participants to engage in real time, high-definition, software supported Telepresence links".[3] The recent work of Whalley (2009, 2010a, 2010b) covers this area and outlines his pioneering work and progress in this field.

### 2.2. Network Approaches

*"There is something magical about gathering together and making sound"*. (Trueman 2007)
It is exciting to be able to perform music with people of other cultures and nationalities. Prior to the advent of high speed internet networks, musician and composers would have to physically meet to collaborate or jam. The internet is now a "faster and more reliable platform for large scale interconnected musical networks." (Weinberg 2005) The speed and capabilities allow for net music interplay many individuals. (Trueman 2007)

Weinberg in his paper *Interconnected Musical Networks* (Weinberg 2005) describes four approaches to online music making. These approaches consider the "level of connectivity among players and the role of the computer in enhancing the interdependent social relations." These describe the ways in which collaborators can interact with each other over the server, which in turn depends on the technical capabilities of the equipment and people involed. Summarised from Weinberg:
The *Server* Approach allows users to send musical information to a server, although it does not allow for users to interact with one another.
The *Bridge* Approach connects distance users and allows them to perform together as if the same space, "providing a technical solution for imitating traditional group collaboration."
The *Shaper* Approach lets users shape and modify music material generated algorithmically by the networks central system. The players can continuously listen and respond to the music that modified by all participants, however, they cannot directly communicate to each other.
The *Construction Kit* Approach morphs together all three of the above approaches in a centralised network. Participants, most likely highly skilled musicians, can contribute the music, manipulate and shape their own and others' players music, and take part in a collective creation.
For the research in question the aim would be to us the Construction Kit approach as this allows the most flexibility between musicians.

When the geographically displaced music makers are joined together by a high speed network, audiences can then be located anywhere around the world. This includes at the physical performance venue(s) and elsewhere streaming the performance to their computer, thus allowing music to reach wider audiences.

[1] Researched by many scholars - including the Journal of the Acoustical Society of America or research listed on www.acoustics.org/booklist.html.
[2] Real space can be defined as the physical present space in which a concert or chamber ensemble would usually play. Virtual space is defined as the space created by the joining of physical spaces via online networks.
[3] http://syneme.ucalgary.ca/tiki-index.php

## 3. THE RESEARCH

The research aim will be to write music for the network music genre. The composer aims to write music with flexible compositional length and structures, rather than predetermine these by conventional notation and composition practise. This will allow for freedom within the works structure and will break away from the "fixed–just-press-play" acousmatic works.[1]

The research will define the role of the composer in creating collaborative music for networked music. History accepted the composer as an authoritative music creator however Stuart Sim argues that in the age of Postmodernism the death of a genius composer has happened. (Sim 1998) With the ease of sampling anyone can create music; this has had a major impact on originality, creativity and ownership. Ten years ago composers most often hand wrote their music scores at the piano - nowadays composers work straight onto computers using notation programmes such as Sibelius with the ability to copy and paste others work. The advent of remixes destroy the "notion of authorship", often removing the composers own presence (sense of self) within their work. (Sim 1998) With networked music and cross culture collaboration questions of ownership will arise as to who came up with what idea and who takes final ownership of the work. In all cases the creative process will be documented to aid this.

All composers want to have their music connect to its audience in some format (Landy 2007) - being able to sense the composer's intention and "soul" in the music is one way to judge the authenticity and effectiveness of a composition. (Emmerson 2007) Members of an audience react to different music as the sound is filtered through their life experience and persona – therefore, maybe, the more "composer personality or being" one hears, the more the audience responds or understands to the music. (Emmerson 2007) The music for this project will attempt to adhere to this aesthetic while having purpose and intention.

The sound resources for this music will be chosen carefully – from samples to instrumentation due to cross cultural collaboration. For example consideration as to the combination of various ethnic instrumentalists and what concerns or intrinsic meaning do they hold and if it is complementary or contradictory. Or, performing a work with bird sounds from different cultures with the addition of a panpipe might just be creating crass world music. To give careful thought to these types of considerations will impact the success, sense and intention of the music.

Depending on the artistic concept of the work, samples will be recorded by the composer from a wide variety of places such as instruments, landscape, nature and people. Which sounds will shape the pieces will depend on the countries involved and the context. How these sounds and instruments are then electronically treated is a major concern. Live sound processing will be primarily done in Ableton Live with a range of spectral plugins. The musicians must be able to interact and change in each performance, and the sound must be able to be shaped and evolve over time.[2]

The music will be flexibly scored, able to be played by varying ensembles in more than one context including a mixture of amateur and virtuosic musicians. This will allow for multiple performances in various sized premises [3] or allow different skilled performers to perform a work.[4]

The performance of Classical music has traditionally been exclusively for highly skilled musicians on Western instruments.[5] Contemporary Classical music also requires virtuoso performers, with most having to learn extended techniques on their instruments to stay current in their field.[6] Amateur musicians are usually self-trained, having had no formal lessons and qualifications on their instruments. Between this lies the Intermediate musician – someone with a great love for music and having had formal lessons but not up to a university performance standard. This project wants to utilize musicians of different capabilities.

The research will also investigate the interchangeability of the instrumentation of works, likening back to the Baroque musical tradition where instrumentation was flexible. Written music works were non-instrument specific and could be performed by which ever instrumentalists turned up that day.

The research is also interested in the multiplicity of interpretations that music can have. How will the music differ from performance to performance? And where is the point of difference that it becomes a separate work? (Benson 2003) Dependant on the interchangeability of instrumentation and different characteristics of (for example) two violins performers it would be interesting to see what qualities of the music are different and unique to each performance. Will changing the instrument forces lessen the authenticity of a work and then what constitutes an authentic performance?

The research will allow the music to have an interchangeable purpose dependant on the performance situation allows for further multiple uses. The reality that

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[2] See the work of (Wishart 1996) and Paine (2002)

[3] For example if space is limited one could leave out the contrabassoon.

[4] For example if it is a piece for a university level string orchestra, but transcribed for high school group yet the electronics and multimedia could still be the same – would it work would it still have the same effect or impact?

[5] For example crowd favourites: Violin Virtuoso Nigel Kennedy or tenor Luciano Pavarotti.

[6] For example *Echange* by Iannis Xenakis, for solo bass clarinet and mixed chamber ensemble, this demands phenomenal control of extended techniques on the soloist's behalf.

concert situations may be vastly different, i.e time frame for performance or technical equipment or the artists performing may have different emotional intentions. As the postmodern view is that there is not one right interpretation or truth, the research will focus on writing music that can exist in many forms at once, all of which may be equally valid.

The music will also tryi to enhance live performers' relationships with the score through improvisation and interaction. "Composers refuse to give up on acoustic forces so easily and are happy not just to coexist but to develop and exploit the often difficult relationship with the technology." (Emmerson in Dean 2009) This is a challenge to create well-crafted works that synthesise acoustic and electronic music.

Advances in technology allow realtime control of electroacoustic music, and institutes such as IRCAM [1] specialise in combining computer and acoustic music. Programmes have been written that interact and react to live inputs then create self-generated music. Roger Dannenberg's research at Carnegie Mellon University[2] enables score following programmes that are active and engaged to *"follow the live performance"* as a human would do. (Emmerson 2009)

There are many interactive systems that are "capable of changing and evolving" through their mutual interaction as described in Paine's theories of Interactivity. (Paine 2002) A recent one being Miroslav Spasov ENATCIVE which is a "multi-modal composing system", which creates content and effects in real time to accompany and interact with a performer. (Spasov 2011)

*"For improvisers and live electronic performers, composition and performance are so strongly integrated that, for some, they are barely separable. Of course, material may be prepared, but may only be finally formed in the moment of performance. Yet again, new versions of very old processes emerge – recycling of materials, reliance on techniques that are known to work".* (Emmerson 2007b)

The research will utilise improvisation to free up the fixed time domain of acousmatic music and to give performers complete control of tempo and expressivity and also, where appropriate, to improvise with the electronic sounds. Emmerson writes how performers can feel "straight-jacketed" by a fixed electronic part and that it surrenders their "precious expressive musical potential" to a machine. (Emmerson 2009)

The research will detail how to perform works at multisite concerts that include physical and virtual spaces.

*"Progressively more advanced media technologies will enhance the sense of telepresence in a wide variety of virtual realities. Rapid advances in both multimedia computer technologies and in high-speed data networks hasten the development of a truly global village, in which our ability to interact with friends, family, and others who share interests similar to our own will no longer be limited by physical proximity. Such "virtual communities" represent one of the most exciting aspects of these developing new media, as they offer individuals a method for participation in, rather than mere observation of, the mediated worlds that surround them."* (Steuer 2007)

As computer technology advances, the world has become a global village, where people's interaction is no longer limited to the physical but can be mediated via computers and networks. This new virtual space has created an online community and allows the chance for musicians and composer to play music together in new ways.

## 4. METHOD

The intellectual and creative core of this research will be the creative and collection of original compositions that synthesises the concerns laid out in section 2 and 3. There will be two methodologies used for this research – one creative and the other analytical. The research will be proven by following a logical progression through the writing of the compositions. To begin small compositions will be written that cover one or many of the eight focus areas. For example, a structured piece for a virtuosic musician in virtual space, or an electronic piece for amateur musicians featuring improvisation.

Each compositional process will be reflexive on the methods applied to future work. The skills and ideas learnt through each composition will inform new methods as well as have a retrospective flow on effect to prior work which could be revisited during the process of research.

*"For my generation it is considered vulgar to have an audience really, really like a piece on the first hearing. Why would we write music except to move people and to be expressive?"* (Benson 2003)

Music will seek to be intentional in its style, meaning and purpose. The music will also seek to engage people intellectually and emotionally while provoking thoughtful responses.

[1] IRCAM = The Institute for Research and Coordination Acoustic/Music. www.ircam.fr.

[2] For more information see www.cs.cmu.edu/~music/ Dannenberg's software is well used – his free audio editor *Audacity* was rated one of the PC World's best 100 products of 2008. http://www.cmu.edu/homepage/innovation/2008/summer/roger-dannenberg.shtml

## 5. CREATIVE OUTPUT

To begin with work has been written:
*Man on the Corner* (2010) Multimedia work for soprano, prepared piano, laptop and visuals. Focuses on: Virtuosic performers, improvisation, electronic and acoustic music, real space.

Short spectral work for orchestra expanding acoustic writing techniques.

*You'll be in all my Tomorrows* (2011) Piano and live electronics work with spectral processing.

*After Sengai* (2011) Live electroacoustic sound for live film by Bevin Yeatman.

### 5.1. Proposed Works

Several small acoustic works for virtuosic performers singularly and combined with amateur performers will be written to develop techniques and ideas.
A suite of electroacoustic music works, each 8-12 minute long which feature techniques in exploring spectral music and further refine control of sound processing. (Sethares 1998)
A small series of improvisation sketches to be performed by acoustic and electronic instruments. This will refine notation methods for improvisation and will test out how much music needs to be notated.
A piece for laptop orchestra that will demonstrate performance value without acoustic instruments.
Real Virtual Space /Networked Music – Music written to be performed in physical and virtual space for online festivals.
These works will be written with the final aim to uniquely synthesise the eight focus areas into large Networked Music projects having discovered appropriate methods within each focus area. The work will be with visual artists, musicians, and other artists from around the globe mainly improvised in real time over high speed networks.

The research focuses will be explored first methodically by following a logical progression through the writing of the compositions. To begin small compositions will be written that cover one or many of the following aspects relating to the research question. For example, a structured piece for a virtuosic musician in virtual space, or an electronic piece for amateur musicians featuring improvisation.
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These works will be written to confirm the skills, technology and resources required for each focus. Each compositional process will be reflexive on the methods applied to future work. The skills and ideas learnt through each composition will inform new methods as well as have a retrospective flow on effect to prior work which could be revisited during the process of research. By reflecting on the success of each piece upon completion it will help to make future informed research decisions.

### 5.2. Performance and Recording

*"Even... composers cannot know their work in the whole fullness of its qualification before its performance… [they] sometimes discover only in performance what exactly they have created."* (Ingarden quoted in Benson 2003)
*"What happens when music-making machines and computers are added to the mix?... The respective roles of concerts and recording have been switched. Recording is the norm and concerts are glorifications of recording."* (Lansky 1990)
Each work will be recorded on CD or DVD, notated appropriately and have detailed analysis of its conception to finish process. The aim of writing these works will be to build a coherent answer to the research questions.
All compositions will have a strong focus on technology and live performance. The technology will be a mix of current software such as Ableton Live, Max for Live and ProTools. The scope of this research will not be to programme new live performance software but to find new ways to combine currently created technology. In all cases an essential focus will be the musical outcome and the technology will serve as means with which to make music.
*"We tend to think of a recording is a documentation of live performance, and perhaps a less than perfect substitute, an illusion, an incomplete and distorted image of reality"*. (Lansky 1990)
The created work will be recorded to serve as a documentation of one interpretation of a live event. It will stand as a "self-contained and self-confirming entity", and will be no substitute for witnessing composer/performer music making interactions via live performance. While the studio will be used as a creative tool for final renders of special audio effects, primarily the work will be live performance.

## 6. CONCLUSION

The intellectual/creative core of this research will be a collection of original compositions that tackle the research aesthetics in a particular way. This work will be documented with the considerations involved with writing this music for physical and virtual space. The initial hypothesis is that the eight aspects of the research will not have equal weighting, some being more important than others and through the unique synthesis and the unusual combination of realtime interactivity, real and virtual space, amateur and professional musicians an original contribution will be made to the genre of Networked Music.

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